Quiz 1 Solutions

Markov Chains and Queuing Systems (EE 621), Spring'26

QUESTION 1

See Fig. 1.

The preceding can be considered to be a renewal reward process, with a new cycle beginning each time a customer arrives to find the door unlocked. This is so because whether or not the arrival finds the server free, the door will become locked for the next t time units and the server will be busy for a time X that is exponentially distributed with rate μ. (If the server is free, X is the service time of the entering customer; if the server is busy, X is the remaining service time of the customer in service.) Since the next cycle will begin at the first arrival epoch after a time t has passed, it follows that

$$E[\text{time of a cycle}] = t + 1/\lambda$$

Let C_1 denote the cost incurred during a cycle due to arrivals finding the door locked. Then, since each arrival in the first t time units of a cycle will result in a cost c, we have

$$E[C_1] = \lambda t c$$

Also, let C_2 denote the cost incurred during a cycle due to an arrival finding the door unlocked but the server busy. Then because a cost K is incurred if the server is still busy a time t after the cycle began and, in addition, the next arrival after that time occurs before the service completion, we see that

$$E[C_2] = Ke^{-\mu t}\frac{\lambda}{\lambda + \mu}$$

Consequently,

$$\text{average cost per unit time} = \frac{\lambda t c + \lambda K e^{-\mu t}/(\lambda + \mu)}{t + 1/\lambda}$$

Fig. 1. The solution for Question 1.

QUESTION 2

For $t \geq 0$, let

$$h(t) = \mathbf{1}_{[0,a](t)} = \begin{cases} 1, & \text{if } 0 \leq t \leq a, \\ 0, & \text{else.} \end{cases}$$

Then $h(t) \geq 0$, $h(t)$ is decreasing, and $\int_0^\infty h(t)dt = a < \infty$. So, $h(t)$ is d.R.i. Now,

$$\int_0^t h(t-x)dm(x) = \int_{t-a}^t dm(x) = m(t) - m(t-a).$$

So, by the key renewal theorem,

$$\lim_{t\to\infty} [m(t) - m(t-a)] = \frac{1}{\mu}\int_0^\infty h(t)dt = \frac{a}{\mu}.$$

Hence, Blackwell's theorem follows.

QUESTION 3

See Fig. 2.

We have that

$$\mathbb{E}(T_M^2) = \mathbb{E}\left\{\left(\sum_{i=1}^\infty Z_i I_{\{M\geq i\}}\right)^2\right\} = \sum_{i=1}^\infty \mathbb{E}\big(Z_i^2 I_{\{M\geq i\}}\big) + 2\sum_{1\leq i<j<\infty} \mathbb{E}\big(Z_i Z_j I_{\{M\geq j\}}\big)$$

since $I_{\{M\geq i\}} I_{\{M\geq j\}} = I_{\{M\geq i\vee j\}}$, where $i \vee j = \max\{i, j\}$. Now

$$\mathbb{E}\big(Z_i^2 I_{\{M\geq i\}}\big) = \mathbb{E}\big(Z_i^2 I_{\{M\leq i-1\}^c}\big) = \mathbb{E}(Z_i^2)\mathbb{P}(M \geq i),$$

since $\{M \leq i-1\}$ is defined in terms of $Z_1, Z_2, \ldots, Z_{i-1}$, and is therefore independent of Z_i. Similarly $\mathbb{E}(Z_i Z_j I_{\{M\geq j\}}) = \mathbb{E}(Z_j)\mathbb{E}(Z_i I_{\{M\geq j\}}) = 0$ if $i < j$. It follows that

$$\mathbb{E}(T_M^2) = \sum_{i=1}^\infty \mathbb{E}(Z_i^2)\mathbb{P}(M \geq i) = \sigma^2 \sum_{i=1}^\infty \mathbb{P}(M \geq i) = \sigma^2 \mathbb{E}(M).$$

Fig. 2. The solution for Question 3.

QUESTION 4

(a) Let X_m be the interarrival time between the mth and $(m-1)$st buses. Then X_m is exponential(λ). Let Y_m be the number of customers entering the mth bus. Then for $k \geq 0$,

$$\begin{aligned} P(Y_m = k) &= \int_0^\infty P\left(Y_m = k | X_m = x\right) \lambda e^{-\lambda x} dx \\ &= \int_0^\infty \frac{e^{-\mu x}(\mu x)^k}{k!} \lambda e^{-\lambda x} dx \\ &= \frac{\lambda \mu^k}{(\mu+\lambda)^{k+1}}. \end{aligned}$$

(b)

$$P\left(Y_m = k | X_m = x\right) = \frac{e^{-\mu x}(\mu x)^k}{k!}.$$

(c) Same as in part (a).

(d) Suppose the $(m-1)$st bus arrives at 10.30 pm and let Y'_m and Y''_m be the numbers of customers that arrive during $[10.30, 11]$ pm and from 11 pm until the mth bus arrives, respectively. Then Y'_m and Y''_m are independent, $Y_m = Y'_m + Y''_m$, and Y'_m is Poisson$\left(\frac{\mu}{2}\right)$ (assuming that the unit of time is an hour).

Y''_m has the PMF calculated in part (a), i.e.,

$$P\left(Y''_m = k\right) = \frac{\lambda \mu^k}{(\mu+\lambda)^{k+1}}.$$

Hence,

$$\begin{aligned} P(Y_m = k) &= \sum_{l=0}^{k} P(Y'_m = l) P(Y''_m = k-l) \\ &= \sum_{l=0}^{k} \frac{e^{-\mu/2}(\mu/2)^l}{l!} \times \frac{\lambda \mu^{k-l}}{(\mu+\lambda)^{k-l+1}} \\ &= e^{-\mu/2} \left(\frac{\lambda}{\lambda+\mu}\right) \left(\frac{\mu}{\lambda+\mu}\right)^k \times \sum_{l=0}^{k} \left\{ \left(\frac{\mu+\lambda}{2}\right)^l \frac{1}{l!} \right\}. \end{aligned}$$

(e) The processes that we see looking backward in time from 2.30 pm are independent Poisson processes with rates λ and μ. Let X' be the time that has elapsed until 2.30 pm since the previous bus arrived and Y' be the number of customers waiting at 2.30 pm. Then X' is exponentially distributed with rate λ and given $X' = x$, Y' is Poisson(μx). Hence, by the same calculation as in part (a), we get that the PMF of Y' is the same as derived in part (a).

(f) Let Y'' be the number of customers that arrive from time 2.30 pm until the next bus arrives. Then Y'' has the same PMF as that found in part (a). Also, with Y' being as in part (e), Y' and Y'' are independent.

Let Y be the number of customers getting on the next bus to arrive after 2.30 pm. Then $Y = Y' + Y''$. So,

$$\begin{aligned} P(Y = k) &= \sum_{l=0}^{k} P(Y' = l)P(Y'' = k - l) \\ &= \sum_{l=0}^{k} \left(\frac{\lambda\mu^l}{(\mu+\lambda)^{l+1}} \times \frac{\lambda\mu^{k-l}}{(\mu+\lambda)^{k-l+1}} \right) \\ &= \sum_{l=0}^{k} \frac{\lambda^2\mu^k}{(\mu+\lambda)^{k+2}} \\ &= \frac{\lambda^2\mu^k(k+1)}{(\mu+\lambda)^{k+2}}. \end{aligned}$$